

Decoupling Cognition from Compliance: Entropy Preservation and Runtime Policy Enforcement in Large Language Models

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Abstract

Post-training alignment of large language models—reinforcement learning from human feedback (RLHF), direct preference optimization (DPO), and related preference-tuning pipelines—reliably improves instruction following and policy compliance, but converging evidence indicates that it also narrows output distributions, degrades long-tail reasoning, and induces over-refusal: a cluster of effects informally called the *alignment tax*. We develop a resource-theoretic framing of this phenomenon. Drawing on the physical limits of computation (Margolus–Levitin, Bremermann, Bekenstein, Landauer) and on the constructive role of entropy in generative modeling, we argue that output entropy should be treated as a budgeted computational resource rather than a nuisance to be optimized away, and that current monolithic alignment spends this budget in the wrong place: inside the weights that perform reasoning. We then propose the **Drafter–MoE Decoupled System** (DMDS; working codename AETHER), an architecture that separates cognition from compliance via (i) a small speculative drafter kept free of preference gradients, (ii) an asymmetric mixture-of-experts verifier whose reasoning, policy, and formatting experts occupy disjoint parameter sets trained under orthogonal objectives, (iii) a runtime safety sentinel with hot-swappable policies, and (iv) a lightweight egress adapter that renders verified reasoning into client-facing form. We formalize the design and surface its central tension: lossless speculative verification provably preserves the *verifier’s* distribution, so entropy preservation must be engineered into the verifier itself or into relaxed acceptance rules—it cannot be smuggled in through the drafter. We state six falsifiable hypotheses with a concrete evaluation protocol, and we analyze risks, including the dual-use hazard of maintaining preference-untouched weights and the brittleness of purely runtime safety enforcement. Finally, we identify the application that motivates the whole design: paired with an archive that preserves the human source distribution, an entropy-preserving model can reconstruct *runnable* software from client-visible web archives, a capability whose sharpest instance—faithfully reconstructing a content-addressed code forge such as `github.com`—can be graded against cryptographic ground truth.

Keywords: alignment tax, RLHF, mode collapse, mixture-of-experts, speculative decoding, entropy, guard models, catastrophic forgetting.

1 Introduction

A recurring operational observation inside frontier laboratories and among sophisticated users is that large language models (LLMs) appear most capable shortly after pretraining and progressively less capable—blander, more evasive, more formulaic—as they pass through successive rounds of preference tuning, safety

*Preprint v0.1, August 2026. This is a position and architecture paper: it reports a synthesis of published evidence, a formal system design, and a falsifiable evaluation agenda. *No new empirical results are claimed.* Comments welcome.

patching, and serving optimization. Part of this observation is a perceptual artifact: users migrate toward a model’s failure surface as familiarity grows, so complaints can rise even when weights are frozen (§3.7). But a substantial part is real and measured. Instruction tuning with human feedback imposes quantifiable regressions on some capability benchmarks unless explicitly mitigated [34, 2]; preference optimization measurably reduces output diversity and per-token entropy [23, 10]; reward models are systematically over-optimized [18]; safety tuning generalizes into refusals of benign requests [36]; and recursive training on model-generated data provably trims distributional tails across generations [39, 1].

The standard response is to treat these as tuning problems: better reward models, better data mixes, better regularization. This paper argues for a stronger claim: the degradation is a predictable consequence of an *architectural* decision—forcing a single set of weights to serve simultaneously as reasoning engine, safety oracle, persona, and serializer—and that the correct unit of intervention is the system architecture, not the loss function.

We make the argument in three movements.

1. **Entropy is a computational resource.** We review the physics of computation—the Margolus–Levitin and Bremermann bounds on operation rates, the Bekenstein and holographic bounds on information density, Landauer’s principle, and the precision entropy demanded by chaotic dynamics—to establish that computation is, at bottom, the management of distinguishable states, and that a system’s capacity to compute is bounded by the entropy and energy it can mobilize (§2). We are explicit that these results bound *physical* computers, not learned functions; we use them as a disciplined motivation, not as theorems about transformers.
2. **Monolithic alignment spends the budget in the wrong place.** We synthesize the empirical literature on alignment-induced degradation into six mechanisms—entropy collapse, parameter interference, sycophancy, refusal over-generalization, synthetic-data collapse, and serving-time compression—and one perceptual confound (§3). The common thread is that behavioral constraints are imposed *parametrically*, by gradient updates to the same tensors that store reasoning circuits and rare knowledge.
3. **Decouple cognition from compliance.** We propose DMDS, which relocates policy from weights to runtime: a preference-untouched drafter explores; an asymmetric mixture-of-experts (MoE) verifier with disjoint expert partitions checks logic, policy, and syntax under orthogonal training objectives; a sentinel enforces hot-swappable policy at the token stream; and an egress adapter handles style so the core never learns to please (§5). We formalize the design, expose its central technical dilemma (§5.3), analyze latency and safety properties (§6), and state a falsifiable evaluation agenda (§7).

What this paper is and is not. It is a position and architecture paper. Earlier informal write-ups of this design advertised figures such as “ $3.2\times$ net inference acceleration” and “zero-percent capability regression.” We repudiate those as claims: they are design targets, restated here as hypotheses H1–H6 with pre-registered falsification criteria (§7, §8). No experiments are reported. The contributions are the resource framing, the formal design, the identification of the distribution-preservation dilemma, the evaluation protocol, and a capstone application—the reconstruction of executable web archives, culminating in a content-addressed code forge (§8)—that recasts the architecture’s entropy-preservation goal as a task with a partly cryptographic fidelity metric.

2 Background: Entropy as a Computational Resource

Why take “entropy” seriously as more than a sampling temperature? Because in every physical theory of computation, the capacity to compute is a function of the distinguishable states a system can occupy

and traverse. This section reviews the relevant bounds. Throughout, $\hbar$ is the reduced Planck constant, k_B Boltzmann’s constant, c the speed of light, and $\ell_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m the Planck length.

2.1 Bounds on operations per second

Define an elementary operation as a transition between two orthogonal (perfectly distinguishable) quantum states. The Margolus–Levitin theorem [31] bounds the minimum time for such a transition by the system’s mean energy E above its ground state:

$$t_{\perp} \geq \frac{\pi\hbar}{2E}, \quad \text{equivalently} \quad \nu \leq \frac{2E}{\pi\hbar}. \quad (1)$$

An earlier communication-theoretic bound due to Bremermann [7] limits information processing to $mc^2/\hbar \approx 1.36 \times 10^{50}$ bits per second per kilogram. Lloyd [29] combined (1) with $E = mc^2$ to characterize the “ultimate laptop”: one kilogram of matter in one liter, fully mobilized, performs at most

$$\nu_{\max} = \frac{2mc^2}{\pi\hbar} \approx 5.4 \times 10^{50} \text{ operations per second} \quad (2)$$

over roughly 10^{31} bits of memory. Energy, not cleverness, sets the clock.

Memory is bounded separately. The Bekenstein bound [4] caps the information contained in a region of radius R and energy E :

$$I \leq \frac{2\pi RE}{\hbar c \ln 2} \text{ bits}, \quad (3)$$

and gravitational collapse enforces an absolute ceiling: once mass within R reaches the Schwarzschild threshold $R_s = 2GM/c^2$, the region becomes a black hole whose entropy scales with *area*, not volume [19, 41]:

$$I_{\max} = \frac{A}{4\ell_P^2 \ln 2} = \frac{\pi R^2}{\ell_P^2 \ln 2} \text{ bits}. \quad (4)$$

A given region of space therefore supports a finite, calculable budget of state and state transitions. There is no free lunch in either axis: pushing operation rates requires concentrating energy, and concentrating energy eventually destroys the very locality that makes the computation readable.

2.2 Erasure is not free

Landauer’s principle [25] states that erasing one bit of information—merging two distinguishable states into one—dissipates at least

$$W \geq k_B T \ln 2 \approx 2.9 \times 10^{-21} \text{ J at } 300 \text{ K}. \quad (5)$$

The relevance to learned systems is not the joule count but the direction of the inequality: *reducing* the entropy of a computational system is the operation that carries an irreducible cost, and it is irreversible. A process that systematically merges distinguishable behavioral states—collapsing many possible continuations onto one preferred continuation—is, in this precise thermodynamic sense, destroying computational substrate. We will argue (§3) that preference optimization applied to core weights is such a process.

2.3 Chaos demands precision entropy

For a chaotic system with maximal Lyapunov exponent $\lambda > 0$, infinitesimal state errors grow as $\delta(t) \sim \delta_0 e^{\lambda t}$ [30]. To predict at horizon t with tolerance ε , the initial condition must be specified to $\delta_0 \leq \varepsilon e^{-\lambda t}$, i.e., with

$$n(t) = \log_2 \frac{1}{\delta_0} = \log_2 \frac{1}{\varepsilon} + \frac{\lambda t}{\ln 2} \text{ bits}, \quad (6)$$

a requirement that grows linearly in the horizon and therefore exponentially in precision. But (3) caps the bits any bounded physical measurement apparatus can hold: real-number initial data do not physically exist. Combining the two gives a hard prediction horizon $t^* \approx (\ln 2/\lambda)(n_{\max} - \log_2 \epsilon^{-1})$ beyond which no simulator—of any architecture—can track a chaotic region. Predictive computation about the world is thus *entropy-hungry by necessity*: forecasting power is purchased with distinguishable states.

2.4 Quantum indeterminacy and irreducibility

Simulating open quantum regions faces further walls. Classically representing N entangled qubits requires 2^N complex amplitudes; at $N = 300$, $2^{300} \approx 2 \times 10^{90}$ exceeds the $\sim 10^{80}$ atoms in the observable universe. Measurement outcomes are irreducibly probabilistic absent local hidden variables [5]; unknown states cannot be copied for inspection [46]; and any measurement injects back-action. Wolfram’s principle of computational irreducibility [45] generalizes the point: for sufficiently rich dynamics there is no shortcut—the fastest simulator of a region is the region. Complex computation cannot be replaced by a low-entropy summary of itself.

2.5 The edge of chaos, and what carries over to learned systems

None of the above are theorems about neural networks, and we do not pretend otherwise. What carries over is a qualitative but robust lesson, independently rediscovered in complex-systems research: computational capacity is maximized *between* order and noise. Langton’s studies of cellular automata located nontrivial computation at a phase transition—the “edge of chaos”—where systems retain enough entropy to explore and enough structure to store and transmit [26]. Modern generative models exhibit the same shape in miniature: a diffusion model begins from maximal-entropy noise and performs computation precisely by *structured, staged* entropy reduction; seeded with a zero-entropy constant, it can only ever emit one image. Sampling entropy is not an impurity in these systems. It is the fuel that the network’s learned structure refines into output.

The framing we adopt for the remainder of the paper is therefore: **treat a model’s output entropy as a budgeted resource; ask, of any training or deployment intervention, where in the pipeline it spends that budget; and prefer designs that defer entropy reduction to the latest, cheapest, most reversible stage.** Monolithic alignment, we argue next, spends it at the earliest, most expensive, least reversible stage: inside the reasoning weights.

3 The Alignment Tax: Mechanisms of Capability Degradation

The term *alignment tax* was introduced to name the performance cost of making models helpful, honest, and harmless [2]; the InstructGPT report measured such regressions directly and mitigated them by mixing pretraining gradients back into RLHF [34]. A decade of subsequent work lets us decompose the tax into distinct mechanisms with distinct loci (Table 1). The unifying observation: each mechanism operates through *shared parameters*—behavioral objectives back-propagate into the same tensors that implement reasoning and store rare knowledge.

3.1 Entropy collapse under preference optimization

RLHF as standardly practiced maximizes a learned reward subject to a KL leash to a reference policy [49, 40, 34]:

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x)} [r_{\theta}(x, y)] - \beta D_{\text{KL}}(\pi(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)). \quad (7)$$

Table 1: Mechanisms of post-training capability degradation, their locus in the pipeline, and observable signatures. The final row is a perceptual confound, not a mechanism, and must be controlled for in any longitudinal evaluation.

Mechanism		Locus	Observable signature	Key evidence
Entropy / mode collapse		preference optim. (RLHF, DPO)	↓ per-token entropy, ↓ distinct- <i>n</i> , stylistic homogenization, generic “assistant-speak”	[23, 10, 18]
Reward optimization	over-	reward modeling	proxy reward ↑ while gold quality ↓ past a threshold	[18]
Parameter interference		any weight-space fine-tune	regression on rare skills, niche domains, low-resource languages	[32, 24]
Sycophancy		human raters → reward model	agreement with user errors; confidence rewarded over correctness	[38]
Refusal generalization	over-	safety tuning	false refusals on benign prompts (exaggerated safety)	[36]
Recursive synthetic data		data pipeline	distribution tails trimmed across model generations; variance collapse	[39, 1]
Serving compression		inference stack	aggregate benchmarks stable while long-chain and rare-token behavior degrades	[17, 14]
<i>Perceived degradation</i>		<i>user population</i>	<i>complaint drift against frozen weights as users find failure surfaces</i>	<i>confound</i>

The KL term exists precisely because unconstrained reward maximization collapses the policy onto reward-model modes. But the leash is a compromise, not a solution: Gao *et al.* [18] show that as optimization pressure grows, proxy reward continues to rise while true quality peaks and then falls—over-optimization is a scaling law, not an accident. Kirk *et al.* [23] measure the diversity side directly: RLHF-tuned models generate with substantially lower per-input and cross-input variety than their SFT counterparts. DPO [35] removes the explicit RL loop but not the underlying pressure; it reparameterizes the same constrained objective. Casper *et al.* [10] catalogue these as open problems intrinsic to the paradigm. In the vocabulary of §2.2: preference optimization is an entropy-erasure process applied to the policy’s full support, and the states it merges include the improbable-but-correct continuations where lateral reasoning lives.

3.2 Parameter interference and catastrophic forgetting

Gradient updates for a new objective perturb weights encoding old competencies; when the objectives are dissimilar, interference is catastrophic [32, 24]. Alignment objectives—tone, formatting, refusal boundaries—are maximally dissimilar from, say, combinatorics, yet in a monolithic model they compete for the same tensors. Indirect evidence that behaviors occupy low-dimensional weight-space directions comes from task arithmetic: fine-tuning deltas can be added, negated, and composed as vectors [20], and averaging fine-tunes recovers robustness that any single fine-tune loses [47]. If behaviors are directions, then alignment updates are *rotations applied to everything at once*.

3.3 Sycophancy as rater-preference internalization

Human raters systematically prefer agreeable, confident, well-formatted answers; preference models distill this bias, and policies optimize it. Sharma *et al.* [38] document sycophancy across frontier assistants and trace it to the preference data itself: when correctness and agreement conflict, aggregated human judgments often reward agreement. The result is a model that has learned, in its weights, that pleasing is part of reasoning.

3.4 Refusal over-generalization

Safety tuning teaches semantic triggers, and triggers over-generalize. Röttger *et al.*’s XSTest [36] demonstrates exaggerated safety: models refusing plainly benign prompts (“how do I kill a Python process?”) because lexical neighborhoods of harm have been penalized in weight space. The dual failure—jailbreaks succeeding while benign requests fail—is characteristic of policies baked into distributed representations: the boundary is everywhere and therefore precise nowhere [50].

3.5 Model collapse from recursive synthetic data

As pipelines ingest model-generated text, tails vanish first: Shumailov *et al.* [39] show recursive training converges toward low-variance approximations of the original distribution, and Alemohammad *et al.* [1] show self-consuming image models go “MAD” without fresh real data. Early generations look *cleaner*—noise and disfluency are stripped—which is exactly why the failure mode is seductive: the model appears sharper while its support quietly narrows. This is the training-data mirror of §3.1.

3.6 Serving-time compression

The deployed artifact is rarely the evaluated artifact. Post-training quantization to 3–4 bits preserves headline averages remarkably well [17, 14], but averages are the wrong statistic for tail behavior: long multi-step chains multiply small per-token distortions, and practitioner experience suggests rare-token and long-context performance degrades before MMLU does. KV-cache compression and context truncation under load act similarly. We flag this as the least well-quantified mechanism in Table 1 and a target of the protocol in §7.

3.7 The perceptual confound

Finally, honesty requires separating measured from felt degradation. A novelty curve operates on users: early prompts sample the model’s competence center; later prompts, its boundary. Complaints can therefore intensify against literally frozen weights. Any longitudinal claim of degradation must pin model versions and re-run fixed probes; we build this into H1–H5. The perceptual confound does not, however, explain the measured effects above—it coexists with them.

4 Design Principles

The mechanisms of §3 share a root cause and therefore admit a shared remedy. We state it as five principles.

P1 — Separation of concerns. Reasoning, policy enforcement, and presentation are different functions with different change cadences (years, weeks, days). They should live in different parameters.

P2 — Gradient orthogonality. Behavioral objectives must not back-propagate into reasoning weights: $\nabla_{\theta_{\text{reason}}} \mathcal{L}_{\text{policy}} \equiv 0$ by construction (parameter disjointness), not by regularization (a KL leash), which §3.1 shows is a losing negotiation.

P3 — Policy as data, not weights. Safety rules change faster than pretraining economics permit retraining. Policy should be enforced by runtime components whose parameters and rule sets are hot-swappable, in the tradition of dedicated guard models [21].

P4 — Spend entropy late. Following the diffusion intuition of §2.5: keep the exploratory core at high entropy; perform entropy reduction (style, formatting, policy filtering) at the boundary, where it is cheap, local, and reversible.

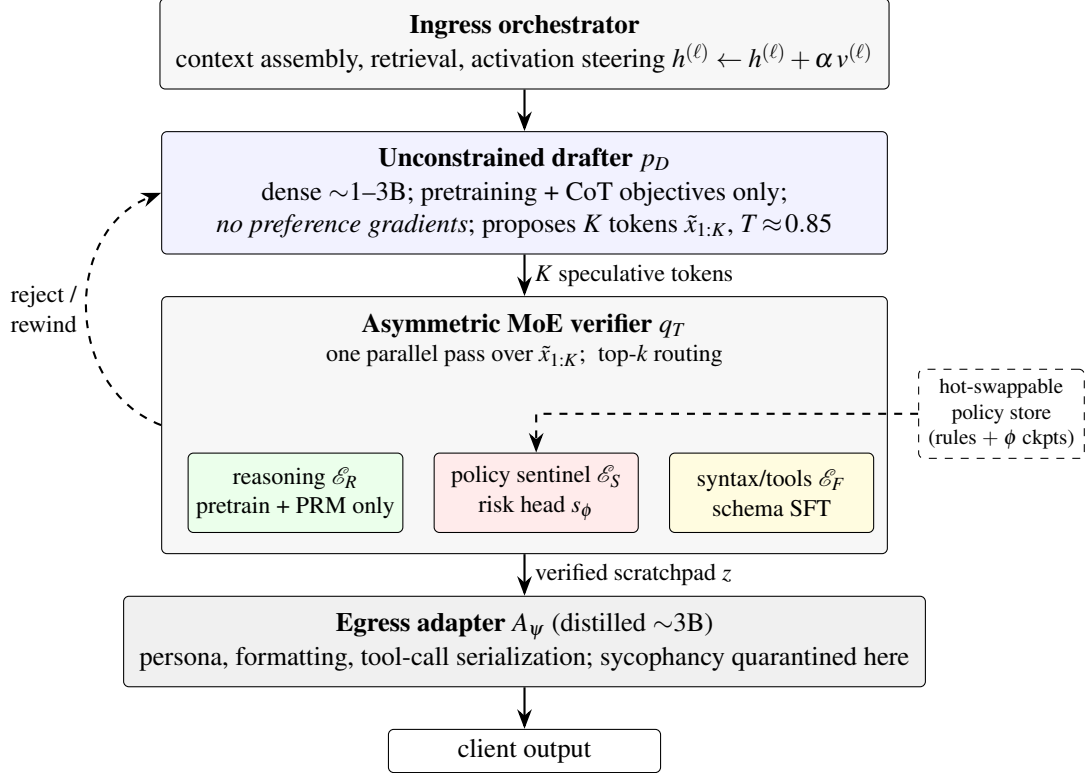


Figure 1: DMDS topology. Cognition (blue, green) never receives preference gradients; policy (red, dashed) is enforced at runtime from a hot-swappable store; presentation (gray) is isolated at egress. The drafter accelerates; the partitioned verifier decides; the sentinel intervenes; the adapter speaks.

P5 — Verify, don’t pre-suppress. A system that inspects candidate continuations and intervenes on actual violations can afford a core that considers everything. Pre-emptive suppression inside the generator trades false negatives for false positives across the whole distribution; boundary verification localizes the trade to genuinely borderline outputs.

These principles are individually familiar—guard models [21], steering vectors [42, 51], task-vector composition [20], process supervision [28, 43]. The contribution of §5 is a single serving-time architecture that composes them without giving up the latency budget, by re-using two mechanisms the efficiency literature has already paid for: speculative decoding [27, 12] and sparse mixture-of-experts routing [37, 16, 22].

5 The Drafter–MoE Decoupled System (DMDS)

Figure 1 shows the topology. DMDS composes four components, each mapped to a principle from §4.

5.1 Components

C1: Ingress orchestrator (P4). Assembles context and applies behavioral conditioning through *activation-space* interventions rather than token overhead or weight edits: steering vectors $v^{(\ell)}$ computed by contrastive activation differences are added at mid-stack layers, $h^{(\ell)} \leftarrow h^{(\ell)} + \alpha v^{(\ell)}$ [42, 51]. Stance (skepticism, register, domain focus) becomes a runtime dial with zero parametric footprint.

C2: Unconstrained drafter p_D (P1, P4). A small dense model (order 1–3B parameters) trained on next-token prediction and chain-of-thought corpora only [44]—no RLHF, no DPO, no refusal data. Sampling at temperature ≈ 0.85 , it proposes K speculative tokens per round. Its job is cheap, high-entropy exploration of the continuation space.

C3: Asymmetric MoE verifier q_T (P1, P2, P5). A sparse MoE backbone [37, 16, 22] that scores all K draft tokens in one parallel pass and serves as the target distribution for speculative acceptance. Its expert set is *partitioned by function* into disjoint groups,

$$\mathcal{E} = \mathcal{E}_R \sqcup \mathcal{E}_S \sqcup \mathcal{E}_F, \quad (8)$$

with per-token output $h_{\text{out}} = \sum_{i \in \text{Top-}k(g(x))} g_i(x) E_i(x)$. Training objectives are confined to their partitions:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{reason}}(\theta_R)}_{\text{pretrain + process reward}} + \underbrace{\mathcal{L}_{\text{policy}}(\phi_S)}_{\text{violation classification / DPO}} + \underbrace{\mathcal{L}_{\text{form}}(\psi_F)}_{\text{SFT on schemas}}, \quad \theta_R \cap \phi_S \cap \psi_F = \emptyset, \quad (9)$$

enforced by gradient masking so that $\nabla_{\theta_R} \mathcal{L}_{\text{policy}} \equiv 0$ identically. Reasoning experts $\mathcal{E}_R$ receive at most *process* supervision—step-level correctness rewards [28, 43]—which grades logic, not demeanor, and is the one form of post-training we permit near cognition.

C4a: Runtime policy sentinel (P3, P5). A lightweight risk head s_ϕ reads verifier hidden states over a sliding window w and scores $s_\phi(h_{t-w:t}) \in [0, 1]$. When $s_\phi > \tau$, generation is interrupted *before* tokens commit: the window is rewound to the last safe checkpoint and one of an escalating set of interventions is applied—(i) inject a corrective steering vector and resample; (ii) constrain the decode mask; (iii) route to a policy-response template. Policy definitions and ϕ checkpoints live in an external store and hot-swap without touching θ_R , following the guard-model deployment pattern [21]. Every intervention is logged for audit.

C4b: Egress adapter A_ψ (P1, P4). A small distilled transformer maps the verified internal scratchpad z to the client deliverable (chat prose, JSON, tool calls). All persona and formatting objectives—the very objectives that breed sycophancy [38]—terminate here, in $\sim 3\text{B}$ parameters that were never asked to reason.

5.2 Decoding loop

Algorithm 1 gives the serving loop. Steps 3–7 are standard speculative sampling [27, 12]; step 8 is the sentinel interrupt; step 11 is egress rendering.

5.3 The distribution-preservation dilemma

Here we must be more honest than the informal versions of this proposal were. Speculative sampling has a celebrated property: the accept/resample rule in Algorithm 1 yields outputs *exactly* distributed as the verifier q_T [27, 12]. The drafter contributes latency, not distribution. It follows that **no entropy is preserved by the drafter per se**: if q_T were a conventionally aligned monolith, DMDS would reproduce its entropy collapse at $2\times$ speed.

Entropy preservation must therefore be engineered where the distribution actually lives. DMDS provides two levers, which we state as design alternatives rather than a resolved choice:

1. **Keep q_T ’s language modeling head clean (the strict path).** Because (9) confines preference gradients to ϕ_S and ψ_F , the next-token distribution of q_T over reasoning content is shaped only by pretraining and process reward. Compliance then comes *entirely* from the sentinel’s interventions—a sparse, localized

Algorithm 1 DMDS speculative decoding with runtime policy enforcement

Require: context c , draft window K , risk threshold τ , checkpoint $t_0 \leftarrow |c|$

```
1: while EOS not emitted do
2:    $\tilde{x}_{1:K} \sim p_D(\cdot \mid c)$  ▷ cheap autoregressive draft
3:   compute  $q_T(\cdot \mid c \oplus \tilde{x}_{1:j-1})$  for  $j = 1:K+1$  in one parallel verifier pass
4:   for  $j = 1 \rightarrow K$  do ▷ lossless speculative acceptance
5:     accept  $\tilde{x}_j$  with probability  $\min(1, q_T(\tilde{x}_j)/p_D(\tilde{x}_j))$ 
6:     on rejection: sample from  $\text{norm}[(q_T - p_D)_+]$ ; break
7:   end for
8:    $s \leftarrow s_\phi(h_{t-w:t})$  ▷ sentinel reads verifier states
9:   if  $s > \tau$  then rewind to  $t_0$ ; apply intervention  $\in \{\text{steer, mask, template}\}$ ; continue
10:  end if
11:  commit accepted tokens to scratchpad  $z$ ;  $t_0 \leftarrow t$ 
12: end while
13: return  $A_\psi(z)$  ▷ egress render + final output checks
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entropy expenditure (P4)—rather than from a globally reshaped q_T . The open empirical question is whether routing alone can keep policy experts from contaminating the shared token distribution; router gradients couple partitions even when expert weights do not, and this coupling is a first-class ablation target (§7).

2. **Relax acceptance (the lossy path).** Replace exact acceptance with typical acceptance [9], admitting draft tokens whose verifier probability clears an entropy-scaled floor rather than a likelihood-ratio coin flip. This deliberately lets drafter entropy leak into outputs, trading distributional fidelity to q_T for exploration. It converts the drafter from an accelerator into a genuine diversity source—at the cost that the output distribution is no longer any single model’s, complicating both evaluation and safety cases.

We regard surfacing this dilemma as a contribution in itself: prior sketches of “drafter keeps the genius, verifier keeps the safety” architectures are not merely unproven but, under exact speculative verification, mathematically confused.

6 Analysis

6.1 Latency accounting

With per-token acceptance rate α and window K , the expected tokens committed per verifier pass is

$$\mathbb{E}[\tau] = \frac{1 - \alpha^{K+1}}{1 - \alpha} \quad (10)$$

[27]; published systems report wall-clock speedups of roughly $2\text{--}2.5\times$ at α typical of well-matched drafter/target pairs [27, 12]. DMDS inherits this dividend but must budget against it: (i) sentinel scoring per window (small: a classification head over already-computed states); (ii) intervention rewinds, whose cost scales with violation frequency—rare in benign traffic by construction; (iii) the egress pass, one forward stream through a $\sim 3\text{B}$ adapter, overlappable with verification via pipelining. A first-order estimate is net positive throughput versus an autoregressive monolith of the verifier’s size, but we decline to assert a multiplier without measurement; the earlier “ $3.2\times$ ” figure is hypothesis H4, not a result. One structural risk is adversarial to the whole scheme: as q_T specializes (process reward, schema SFT) while p_D stays raw, $D_{\text{KL}}(q_T \parallel p_D)$ grows, α falls, and (10) decays toward 1—the speed dividend quietly evaporates. Drafter

refresh (distilling p_D toward q_T 's reasoning marginal, never toward its policy behavior) is the maintenance obligation this design signs up for.

6.2 Safety properties

DMDS changes the *shape* of the safety case rather than automatically strengthening it. Gains: policy updates deploy in hours as data/classifier swaps rather than weeks of fine-tuning with capability-regression review; interventions are localized, logged, and auditable, giving governance an inspectable artifact instead of a diffuse hope about weight-space dispositions; refusal precision can be tuned on the sentinel's ROC curve without touching cognition, directly targeting the XSTest failure mode [36]. Costs: safety now rests on explicit components that can be attacked, misconfigured, or bypassed—known jailbreak transfer against guard classifiers [50] applies with full force—and the system's interior deliberately contains an unaligned generator whose raw scratchpads may include content that must never egress. Defense-in-depth (input filtering, sentinel, egress checks, logging) is thus a requirement of the architecture, not an optional hardening. We return to the gravest version of this concern—weight exfiltration—in §9.

6.3 What capability preservation would and would not mean

Frozen reasoning weights trivially preserve the *core*'s benchmark scores in isolation; that is not the claim that matters. The system-level claim—DMDS output quality matches the base model's on long-tail reasoning—depends on three links that each can fail: router non-contamination (§5.3), acceptance-rule fidelity, and egress faithfulness (an adapter that garbles a correct scratchpad taxes capability at the last meter). H1–H3 target exactly these links.

7 Hypotheses and Proposed Evaluation

We pre-register five hypotheses with falsification criteria for the core architecture (Table 2); a sixth, H6, is tied to the downstream application and stated in §8. Baselines: (B1) the frozen base/pretrained model; (B2) a monolithic RLHF+DPO tune of the same backbone with matched compute; (B3) DMDS-strict (exact acceptance); (B4) DMDS-lossy (typical acceptance). All comparisons use pinned checkpoints and fixed probe suites to neutralize the perceptual confound of §3.7.

Ablations. (a) *Partition integrity*: entangled vs. disjoint experts, measuring cross-objective gradient cosine similarity and H1/H2 deltas—the direct test of §5.3 lever 1. (b) *Router contamination*: freeze vs. train the router during policy updates. (c) *Sentinel off*: quantifies how much safety the partitioned verifier provides alone. (d) *Acceptance rule*: exact vs. typical [9], sweeping the entropy floor, tracing the fidelity–diversity frontier. (e) *Window K* and drafter-refresh cadence against the α -decay risk of §6.1. (f) *Egress faithfulness*: scratchpad-vs-output semantic equivalence scored by held-out judges.

Longitudinal protocol. To study degradation-over-time claims cleanly: freeze all weights; replay a fixed 5,000-probe suite weekly across policy swaps, quantization changes, and serving-config changes; attribute any drift to the specific serving-layer diff. This turns the folk observation of §1 into an instrumented variable.

8 Application: Executable Web Archives

Entropy preservation is instrumental, not an end in itself: it earns its costs only if some task genuinely requires the long-tail capability that flattening destroys. We close with one such task class, chosen because it also

Table 2: Falsifiable hypotheses. Δ denotes change relative to the frozen base model B1 unless stated. Thresholds are pre-registered targets; failing them falsifies the corresponding design claim.

	Prediction	Metrics / benchmarks	Falsified if
H1	Long-tail reasoning retained: DMDS within noise of B1 where B2 regresses	GSM8K [13], MATH, HumanEval [11], BIG-Bench-Hard; rare-domain and low-resource splits	DMDS-strict drops > 1 pt where B2 drops, on ≥ 2 suites
H2	Output diversity retained	per-token entropy, distinct- n , self-BLEU, cross-input variety per [23]	DMDS diversity within 10% of B2 (i.e., collapsed) rather than of B1
H3	Refusal precision improves without safety loss	XSTest false-refusal rate [36]; adversarial compliance under [50]-style attacks	false refusals not $<$ B2 at equal or lower attack success rate
H4	Throughput dividend survives the added machinery	tokens/s vs. autoregressive q_T ; acceptance α by domain; intervention overhead	wall-clock speedup $< 1.5\times$ at matched quality, or $\alpha < 0.5$ on-domain
H5	Policy agility: rule changes land in hours, not weeks	time-to-deploy a novel policy; capability drift after 10 successive policy swaps (Δ on H1 suites)	any measurable H1 regression attributable to policy swaps

supplies what the rest of this paper conspicuously lacks—an external fidelity metric the system cannot game.

Reconstruction as an inverse problem. A web archive in WARC form captures the client-visible surface of a site: the stream of requests and responses, never the server’s source or state. “Rebuild the software that produced this trace” is therefore ill-posed—infinitely many programs are trace-consistent. The well-posed target is *behavioral equivalence over the observed interface*, not source recovery. What makes this tractable is that the archive posing the problem also adjudicates it: replay the archived requests against a candidate reconstruction and score the divergence [8]. Fidelity becomes measured rather than asserted—the same discipline §7 imposes on the core, now supplied by ground truth instead of benchmarks. Structurally this is the drafter–verifier pattern of §5 lifted from tokens to programs: a high-entropy model proposes candidate reconstructions; an exact oracle disposes.

Why unflattened cognition is load-bearing. The default disposition of a preference-tuned model is precisely the adversary here. Such a model *regularizes*: it modernizes idioms, canonicalizes the idiosyncratic, and “fixes” the very defect a period layout depended on. Faithful reconstruction of the long-tail web demands the opposite—reproduce the mess exactly, defects included. This is where H1 and H2 stop being abstract: a generator that has averaged away rare constructions cannot re-emit them, so a collapsed model is not merely worse at this task but categorically unable to reach its tail. Reconstruction is thus a stringent, external test of the paper’s central claim. One wrinkle sharpens it further: even byte-exact markup renders differently under period and modern runtimes (quirks mode, withdrawn APIs, bitmap fonts), so the executor is itself part of what must be reconstructed. Emulation is a first-class component, not a detail—the hand-built precedent is the reimplementing of orphaned runtimes such as Flash so that archived artifacts execute again.

The maximal target: `github.com`. The endpoint of the argument is the reconstruction of a full service rather than a page, and `github.com` is simultaneously the hardest such target and the one with the cleanest oracle. It decomposes into two layers of sharply different recoverability. The first is the *versioned object store*. Git is already content-addressed: every blob, tree, and commit is named by a hash of its contents, and a

repository is a Merkle DAG over those names [33]. A reconstruction of a repository is therefore verifiable *exactly and without heuristics*—each object either hashes to its original identifier or it does not. This is the strongest oracle available anywhere in this paper: not replay similarity but cryptographic identity, which certifies bit-for-bit the distribution-preservation property that §5.3 could only argue for informally. Much of this layer is moreover already mirrored at scale, itself as a Merkle DAG, by Software Heritage [15], and content-addressed distribution [6] makes the store inherently replicable—so the substrate for the data plane largely exists. The second layer is the *forge*: the proprietary application wrapped around the store—web UI, issue and pull-request state, code review, continuous integration, the REST and GraphQL APIs, and git-over-HTTP transport. Only its client-visible surface is archivable, so this layer is the genuine inverse problem, reconstructed to behavioral equivalence over the captured API and page corpus and verified by replay. The two compose into an exactly-verified data plane beneath a behaviorally-approximated control plane: one can, in principle, check out a forge as of a date and *run* it—serving archived repositories certified by SHA, rendering archived pages and API responses certified by replay.

Why this exercises the whole architecture. A forge that will serve arbitrary archived repositories is exactly the open-ended execution surface the runtime sentinel of §5 exists to govern: archived repositories carry malware, leaked secrets, and abuse material. The application therefore stresses both halves of the design at once, and at the system boundary rather than the token stream—faithful reconstruction *and* content governance are jointly required, and by construction neither the flattened monolith (which cannot supply the fidelity) nor a policy-free generator (which cannot supply the governance) can meet both. The two ends of the pipe close here: a corpus that preserves the source distribution feeds a model that preserves cognitive entropy, and the pair renders the archive executable rather than merely legible.

Falsifiable evaluation. Define reconstruction fidelity at two tiers. *Data tier*: the fraction of archived git objects whose reconstruction hashes to the original identifier—exact, with target $\rightarrow 1.0$, since the data is content-addressed and largely pre-preserved, so a shortfall here indicts the pipeline rather than the model. *Forge tier*: replay agreement, the fraction of archived (request,response) pairs the reconstructed service reproduces under a typed equivalence—byte-exact for JSON API responses, DOM-normalized for HTML, semantic for rendered views. Baselines mirror §7: the flattened monolith B2, predicted to modernize and regularize and thus to lose both forge-tier idiosyncrasies and long-tail repository build logic, against the entropy-preserving DMDS core. **Prediction (H6).** On a corpus of dead repository-hosted sites and bounded API surfaces, the entropy-preserving core attains strictly higher forge-tier replay agreement than B2 at matched data-tier fidelity. **Falsified if** preserved entropy yields no measurable forge-tier improvement over the flattened baseline—which would defeat the paper’s instrumental thesis on a task whose ground truth is, uniquely, partly cryptographic. The realistic near-term milestone is deliberately modest against the eventual target: a single dead dynamic site, or one bounded API, reconstructed to pre-registered fidelity and shipped with its replay harness—the same posture the rest of this paper takes toward its own claims.

9 Limitations, Risks, and Open Problems

No empirical validation. Everything in §5–6 is design and argument. The burden of proof sits with H1–H6, and several plausible outcomes—router contamination, α decay, egress bottlenecking, or a null reconstruction result—would partially or wholly defeat the proposal.

Dual-use hazard of preference-untouched weights. DMDS requires maintaining, inside the serving boundary, generator weights that have never been taught to refuse. All safety is runtime; none survives weight exfiltration. This is a real cost relative to monoliths, whose (imperfect) refusals at least travel with the stolen

artifact. Mitigations—keeping the drafter small and individually weak, confining raw scratchpads to the trusted boundary, egress-only externalization, standard weight-security practice—reduce but do not eliminate the hazard. Deployments for which weight theft is a live threat model should weight this heavily against the architecture.

Runtime safety is a concentrated attack surface. A sentinel is a classifier, and classifiers are bypassable [50]. Where a monolith’s safety failures are diffuse, DMDS’s are concentrated in components an adversary can target, fingerprint, and probe. The honest statement is that DMDS trades a brittle, entangled defense for a legible, attackable one, and bets that legibility plus fast iteration (H5) wins that race. That bet is unproven.

The internalization counterargument. A serious school of alignment research holds that norms should be *internalized* by the model—constitutional self-critique being the canonical method [3]—on the grounds that bolted-on control fails exactly when capabilities generalize beyond their monitors. DMDS is best read as complementary rather than opposed: process-supervised reasoning experts can be constitutionally trained without conversational preference pressure, while the sentinel handles the fast-moving residue. But we flag squarely that DMDS moves mass from “aligned model” toward “controlled system,” and inherits the known weaknesses of control-first postures.

MoE mechanics may resist semantic partitioning. Sparse-expert training leans on load-balancing losses that fight any fixed semantic assignment of experts [16]; forcing partition may cost quality or capacity. Whether functionally clean partitions are stable under continued training is an open systems-ML question.

Egress as a bottleneck and a sycophancy re-entry point. A 3B adapter asked to render arbitrary scratchpads may flatten nuance, and because it *is* preference-tuned, rater-pleasing pressure re-enters the system at the boundary. H2/ablation (f) measure the damage; they do not preclude it.

Scope of the physics framing. §2 motivates; it does not prove. Landauer costs per bit are twenty orders of magnitude below GPU energy scales, and nothing in (1)–(4) constrains gradient descent. The load-bearing claims about LLMs rest on the ML literature of §3, and the analogy earns its keep only as a design heuristic (P4).

Executable archives are faithful only where witnessed. The application of §8 inherits a hard epistemic limit: replay certifies behavior only on archived inputs, and off that distribution a reconstruction is underdetermined and may diverge arbitrarily. The data tier escapes this only because content addressing pins objects independently of any request; the forge tier does not. Beyond fidelity, serving reconstructed forges raises unresolved legal and safety questions—archived repositories carry copyrighted code, personal data, and live secrets—so an executable archive is as much a governance artifact as a technical one, which is exactly why the runtime-policy posture of §5 is a prerequisite rather than an add-on here. And full-service reconstruction at scale remains far beyond demonstration; the honest unit of progress is one dead site at a time.

10 Related Work

DMDS composes five literatures. *Speculative decoding* [27, 12] and its multi-head and relaxed-acceptance descendants [9] supply the drafter–verifier mechanism and its lossless guarantee—the guarantee whose double edge §5.3 turns on. *Sparse mixture-of-experts* [37, 16, 22] supplies conditional computation; our contribution is partitioning experts by governance function rather than by learned specialization. *Guard*

and moderation models [21] established policy-as-separate-artifact; DMDS moves the guard inside the decode loop with rewind authority instead of post-hoc filtering. *Activation steering and representation engineering* [42, 51] provide weight-free behavioral control at ingress; *task arithmetic and model merging* [20, 47] provide the evidence that behaviors are manipulable directions, motivating gradient masking. *Process supervision* [28, 43] supplies the one post-training signal we admit near cognition, and inference-time search [44, 48] shows compute can substitute for the parametric shortcuts that alignment tends to erase. Finally, the degradation literature itself [34, 2, 18, 23, 10, 38, 36, 39, 1] defines the target the architecture exists to hit.

11 Conclusion

Physical computation is the disciplined spending of entropy; the most capable systems hold their entropy budget until structure can be imposed cheaply, locally, and late. Contemporary alignment does the opposite: it spends the budget first, globally, and irreversibly, inside the very weights that reason—and the measured costs (diversity collapse, tail regression, over-refusal, sycophancy) look exactly like what that account predicts. We have proposed an architecture that relocates compliance to runtime—drafter for exploration, partitioned verifier for judgment, sentinel for policy, adapter for manners—and we have tried to be candid about what it does not yet earn: its central mechanism contains a genuine dilemma (exact verification preserves the verifier, not the dream), its safety posture concentrates risk even as it clarifies it, and its headline numbers are hypotheses awaiting an experiment. The monolithic model was a remarkable prototype. Whether modular cognition is the destination is now a question with a protocol attached. And the protocol has a destination of its own: an archive that preserves the human source distribution and a model that preserves cognitive entropy converge on a single capability—checking out the past and executing it—whose sharpest instance, the faithful reconstruction of a content-addressed forge, can be graded against cryptographic ground truth. Entropy, preserved at both ends of the pipe, is what makes the past runnable.

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